

DO NOT WRITE ON TEST



IT Concepts
~OPEN EVENT~
(391)

State 2024

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. Which port is commonly used for SMTP communication?
 - A. 21
 - B. 25
 - C. 53
 - D. 80
2. Which protocol should be used for remote terminal access?
 - A. FTP
 - B. Telnet
 - C. SSH
 - D. SMTP
3. Which Linux command is used to create a new directory?
 - A. mkdir
 - B. rmdir
 - C. ls
 - D. rm
4. Which Linux command is used to display the current working directory?
 - A. pwd
 - B. cd
 - C. ls
 - D. rm
5. Which component of Windows 10 or 11 is responsible for managing user accounts?
 - A. Security settings
 - B. Control Panel
 - C. Task Manager
 - D. Local Users and Groups
6. Which protocol is used for sending email messages?
 - A. SMTP
 - B. POP3
 - C. IMAP
 - D. HTTP
7. What is the purpose of a loop in programming?
 - A. To execute a set of statements repeatedly
 - B. To break the execution of a program
 - C. To define a new variable
 - D. To compare two values

8. What is the difference between a variable and a constant in programming?
 - A. Variables can be changed, while constants remain constant
 - B. Variables are used for calculations, while constants are used for storage
 - C. Variables are global, while constants are local
 - D. Variables are used in object-oriented programming, while constants are used in procedural programming
9. What is the purpose of a Creative Commons license?
 - A. To protect intellectual property from unauthorized use
 - B. To promote the sharing and distribution of creative works
 - C. To enforce copyright restrictions on digital media
 - D. To limit access to copyrighted material
10. What is the purpose of an API (Application Programming Interface)?
 - A. To provide a graphical user interface for an application
 - B. To enable communication and interaction between different software applications
 - C. To define the structure and layout of a database
 - D. To encrypt and decrypt sensitive data
11. What is the purpose of a firewall in network security?
 - A. To block unauthorized access to a network
 - B. To speed up network communication
 - C. To encrypt network traffic
 - D. To connect different networks together
12. Which WiFi security protocol is considered the most secure?
 - A. WEP
 - B. WPA
 - C. WPA2
 - D. WPA3
13. Which programming language is primarily used for game development?
 - A. C++
 - B. Python
 - C. PHP
 - D. Ruby
14. Which WiFi standard introduced multiple-input and multiple-output (MIMO) technology?
 - A. 802.11a
 - B. 802.11b
 - C. 802.11g
 - D. 802.11n

15. Which WiFi frequency band provides better range but lower data transfer speeds?
 - A. 2.4 GHz
 - B. 5 GHz
 - C. 2.4 GHz and 5 GHz
 - D. 900 MHz
16. What is the purpose of an access point in a WiFi network?
 - A. To connect multiple wireless devices to a network
 - B. To assign IP addresses to devices on a network
 - C. To filter and block unwanted network traffic
 - D. To secure the network from unauthorized access
17. Which WiFi security protocol should be used for maximum compatibility with older devices?
 - A. WEP
 - B. WPA
 - C. WPA2
 - D. WPA3
18. What is the purpose of a VPN (Virtual Private Network)?
 - A. To provide a graphical user interface for an application
 - B. To enable communication and interaction between different software applications
 - C. To secure and encrypt network traffic over an untrusted network
 - D. To connect multiple networks together
19. What is the purpose of a package manager in programming?
 - A. To manage and track changes to source code files
 - B. To automatically test software applications
 - C. To install, update, and remove software libraries and dependencies
 - D. To optimize code execution speed
20. Which type of memory is volatile and loses its data when the power is turned off?
 - A. ROM
 - B. RAM
 - C. Cache
 - D. Hard drive
21. What is the purpose of a graphics card?
 - A. To provide additional storage capacity
 - B. To connect external devices to the computer
 - C. To enhance the performance of the CPU
 - D. To render and display images on a monitor

22. Which type of cable is commonly used to connect a computer to a local area network (LAN)?
- A. USB
 - B. HDMI
 - C. Ethernet
 - D. VGA
23. Which component is responsible for converting digital signals to analog signals for communication over telephone lines?
- A. Modem
 - B. Router
 - C. Switch
 - D. Firewall
24. Which component is responsible for providing temporary storage for frequently accessed data to improve system performance?
- A. Hard drive
 - B. CPU
 - C. Cache
 - D. Optical drive
25. Which of the following is an example of a solid-state storage device?
- A. Hard drive
 - B. DVD drive
 - C. Flash drive
 - D. Optical drive
26. What is the purpose of a UPS (Uninterruptible Power Supply)?
- A. To provide additional cooling for the CPU
 - B. To protect the computer from power surges and outages
 - C. To improve network connectivity
 - D. To store and backup data
27. What is the purpose of a heat sink in a computer?
- A. To store data temporarily
 - B. To provide power to the CPU
 - C. To cool down the CPU
 - D. To enhance network connectivity
28. What is the purpose of an expansion slot in a computer?
- A. To store the operating system
 - B. To connect the computer to the internet
 - C. To add additional hardware components
 - D. To provide power to the computer

29. What is the purpose of a network switch?
- A. To connect multiple devices to a network
 - B. To protect the computer from viruses
 - C. To store and manage files on a network
 - D. To improve the computer's processing speed
30. What is the purpose of a CD/DVD drive in a computer?
- A. To provide power to the computer
 - B. To connect the computer to external devices
 - C. To store and retrieve data from optical discs
 - D. To enhance graphics performance
31. Which type of software allows users to browse the web and access websites?
- A. Spreadsheet software
 - B. Antivirus software
 - C. Web browser
 - D. Graphic design software
32. Which encryption algorithm is commonly used for secure data transmission over the internet?
- A. AES
 - B. RSA
 - C. DES
 - D. SHA
33. In asymmetric encryption, how many keys are used for encryption and decryption?
- A. One key
 - B. Two keys
 - C. Three keys
 - D. Four keys
34. Which of the following is a common cryptographic hash function?
- A. MD5
 - B. SHA-256
 - C. RC4
 - D. Triple DES
35. What is the term for a cryptographic technique that hides data within other data?
- A. Steganography
 - B. Substitution
 - C. Permutation
 - D. Key exchange

36. Which law is designed to protect individuals' personal data and privacy rights in the European Union?
- A. Data Protection Act (DPA)
 - B. Personal Information Protection and Electronic Documents Act (PIPEDA)
 - C. General Data Protection Regulation (GDPR)
 - D. Computer Fraud and Abuse Act (CFAA)
37. What is the role of an ethical hacker?
- A. To hack into systems for personal gain
 - B. To exploit vulnerabilities for malicious purposes
 - C. To identify and report security weaknesses to improve security
 - D. To sell stolen data on the dark web
38. What is the primary goal of ergonomics in the context of Information Technology?
- A. Maximizing profits for IT companies
 - B. Enhancing user experience and productivity while reducing physical strain
 - C. Designing aesthetically pleasing products
 - D. Decreasing energy consumption in IT devices
39. How can Information Technology contribute to environmental sustainability?
- A. By increasing the production of single-use plastic products
 - B. By encouraging excessive printing of documents
 - C. By promoting the use of energy-efficient devices and servers
 - D. By creating software that consumes more system resources
40. Which component of a printer is responsible for transferring the toner onto the paper during the printing process?
- A. Fuser
 - B. Toner Cartridge
 - C. Imaging Drum
 - D. Paper Tray
41. Which printer technology relies on a heated printhead to create images on heat-sensitive paper?
- A. Inkjet Printer
 - B. Laser Printer
 - C. Dot Matrix Printer
 - D. Thermal Printer
42. What is the primary advantage of using a wireless printer?
- A. Faster printing speed
 - B. Lower cost of consumables
 - C. No need for physical connections, allowing printing from multiple devices
 - D. Higher resolution output

43. Which graphics file format is best suited for storing high-quality images with complex details and photographs?
- A. JPEG
 - B. GIF
 - C. TIFF
 - D. PNG
44. What is the term used for a single point in a digital image that contains color information?
- A. Pixel
 - B. Resolution
 - C. Aspect Ratio
 - D. Vector
45. What does the principle of "informed consent" mean in the context of Information Technology?
- A. Using technology without any restrictions
 - B. Providing personal information without awareness
 - C. Obtaining explicit permission before collecting and using user data
 - D. Sharing sensitive data with unauthorized parties
46. What is the significance of the principle of "intellectual property rights" in IT ethics?
- A. Encouraging the use of copyrighted materials without permission
 - B. Protecting the rights of software developers and content creators
 - C. Ignoring patents and trademarks related to IT products and services
 - D. Promoting open-source software exclusively
47. What is the primary purpose of a "code of ethics" in the IT profession?
- A. To restrict competition among IT professionals
 - B. To regulate software licenses and usage
 - C. To provide guidance and standards for ethical behavior in the industry
 - D. To favor one technology vendor over others
48. Which operating system is developed by Apple Inc. and is commonly used in Mac computers?
- A. Windows
 - B. Linux
 - C. macOS
 - D. Android
49. Which file system is commonly used in Windows operating systems?
- A. NTFS (New Technology File System)
 - B. HFS+ (Hierarchical File System Plus)
 - C. ext4 (Fourth Extended File System)
 - D. APFS (Apple File System)

50. Which operating system is open-source and based on the Linux kernel?
- A. Windows
 - B. Ubuntu
 - C. macOS
 - D. Android